THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 36	the rule gives 36
2	**(a)** 11	the rule gives 11
3	**(b)** 64	8×8
4	**(a)** 50	$7^2, 8^2, 9^2$ are squares
5	**(c)** 55	$21 + 34$
6	**(a)** adding a ‘speed bump’ to each line segment	shape pattern
7	**(d)** the ‘all 1’s’ sequence	constant sequence
8	**(b)** pattern rule	describes sequence
9	**(b)** 45	$1 + 2 + \cdots + 9 = 45$
10	**(d)** 144	$n^2 = 144$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1, 4, 9, 16, 25.	1+1
12	1, 3, 6, 10, 15.	1+1
13	Dots equal to these numbers can be arranged to form a perfect square.	1+1
14	Sum of the first 5 odd numbers is $5^2 = 25$.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add one L-shaped layer of odd number of dots to a square each time; 1, 1+3 = 4, 1+3+5 = 9 (dots form bigger squares).	1½+1½
16	Dots equal to each number can be arranged in a triangle with 1, 2, 3 … dots in rows (1½); each new number adds one more row (1½).	1½+1½
17	Each term is the sum of the two before it (1); 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 (2).	1+2
	OR: Triangular number: 55 (1); pairs $(1 + 10), (2 + 9) \ldots$ make 5 pairs of 11 (2).	1+2
18	Add 3 each time (1); 16, 19, 22 (2).	1+2
	OR: The next shape has 16 dots (1); a 4 by 4 grid (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1, 3, 6, 10, 15, 21, 28, 36 (2). (ii) e.g. 10+15 = 25 (1.5). (iii) Rows of 1, 2, 3, 4 dots (1.5).	2+1½+1½
	OR: (i) 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 (2). (ii) L-layers forming 4 by 4 square (1.5). (iii) $12^2 = 144$ (1.5).	2+1½+1½
20	(i) Each term is the sum of the previous two (1.5). (ii) 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233 (2). (iii) Petals, spirals of sunflower seeds (1.5).	1½+2+1½
	OR: (i) 1, 8, 27, 64, 125, 216; 1, 2, 4, 8, 16, 32 (2). (ii) Powers of 2 grow faster later on (1.5). (iii) 1000 and 512 (1.5).	2+1½+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 21	the rule gives 21
2	**(a)** 12	the rule gives 12
3	**(d)** 125	$5 \times 5 \times 5$
4	**(c)** 28	$1 + 2 + \cdots + 7 = 28$
5	**(c)** 3, 4, 5, 6, …	one more each time
6	**(d)** square numbers	odd sums
7	**(a)** 200	2×100
8	**(d)** multiples of 3	$3 \times n$
9	**(a)** 66	$1 + 2 + \cdots + 11 = 66$
10	**(d)** 225	$n^2 = 225$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A list of numbers or shapes in a particular order following a rule; e.g. 2, 4, 6, 8.	1+1
12	1, 2, 4, 8, 16, 32.	1+1
13	$7^3 = 343$.	1+1
14	13 and 21 (each term is the sum of the previous two).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1, 8, 27, 64, … (1); each is a number multiplied by itself thrice (1); they can be shown as cubes of unit blocks (1).	1+1+1
16	Each new odd number adds a layer around the earlier square (1.5); so squares keep growing (1.5).	1½+1½
17	Formed by adding counting numbers; two consecutive ones add to a square; shown by dot triangles; 1, 3, 6, 10, … (any three with one more).	1½+1½
	OR: A series of shapes following a pattern (1); e.g. regular polygons with 3, 4, 5, … sides (2).	1+2
18	Triangular numbers (1.5); next is 15 (1.5).	1½+1½
	OR: Multiples of 3: $3 \times 20 = 60$ (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 7 (1). (ii) 28 (1). (iii) Triangular numbers (1). (iv) 8 rows (2).	1+1+1+2
	OR: (i) 162, 486 (2). (ii) Multiply by 3 (1.5). (iii) $2 \times 3^6 = 1458$ (1.5).	2+1½+1½
20	(i) 1, 3, 6, 10, 15, 21 (2). (ii) Triangular numbers (1). (iii) 45 (2).	2+1+2
	OR: (i) Shapes following a rule (1). (ii) 1, 4, 9 squares (2). (iii) 25 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.